

THE WILD BLUE YONDER

The quest for control of the skies and the sea

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He who controls the skies controls the battlefield. Whoever said that had his head screwed on right. In the quest to find that proverbial high-ground, air supremacy is the highest ground there is (barring satellites of course). Air supremacy grants you full freedom of movement, effective reconnaissance and the ability to lay waste to ground targets with relative impunity.

Helicopters are particularly suited for rapid troop deployments and as mentioned earlier, infantry are key to any military operation and CAS (Close Air Support) is the best way to support them. CAS can be via bombing designated targets, calling in and ranging in effective artillery barrages, spotting the enemy and providing transport and evac as and when required.

To achieve air superiority, a number of means are used. Air superiority fighters are the most exciting example, fast and nimble jets that are designed to make mince-meat of the opposition. When you're on home ground, ground-based radar and missile systems are your best bet, complemented by your own air-superiority fighters of course.

Every nation has its own ideas when it comes to achieving air superiority. Here are some interesting facts.

STEALTH THE DARK SECRET

Stealth is the holy grail of the American military, in fact, it might be the Holy Grail for every military the world over. The dream is to have an aircraft that can sneak behind enemy lines, accomplish its objectives and sneak out. Nobody actually

wants a dog fight. Why should you fight to the death when all you would really like to see is a blip on a scope and, at the press of a button, it's gone. This is the dream. Unfortunately, it might remain a dream for a long time yet.

The problem here is the lack of a completely reliable and unique friend or foe identification system. Essentially, the way things stand, you don't know if that blip on your radar is friend or foe. Current RoE (Rules of Engagement) prevent most pilots from firing at targets unless they've been visually identified and even if you assume stealth works as advertised (which is questionable), it's not effective at visual range.

The second point against stealth is the fact that it is, in many ways, a flawed concept. The primary objective of a stealth

aircraft is for it to appear invisible to enemy radar. Unfortunately, to achieve stealth, most aircraft need to compromise on speed and agility to accommodate the extra weight of the radar-absorbent paint and reduce the infra-red signature of the engines. Imagine if you had an entire fleet of such aircraft, aircraft that you imagined to be invisible and invulnerable, and you launched said fleet against a rival. Suddenly, all your aircraft have been engaged and are falling like flies! What's gone so catastrophically wrong?

The Achilles heel of stealth technology was apparently discovered in the 1940s. The primitive radar systems used by the British in WWII used radio waves operating at long wavelengths and without going into the technical details, these long wavelengths are perfect for detecting stealth aircraft. The Russians were quick to pick up on this and they have been, for more than three decades, building and perfecting anti-stealth radar systems which they've been handing over to anyone who asks for it.

Case in point is the shooting down of the American F-117 Nighthawk by the Serbian armed forces. How did they do it? A little birdie suggested a few modifications to their "obsolete" Soviet radar equipment. Those changes painted the F-117 on its return from a bombing run and taking it down was just a matter of course. A pilot downed behind enemy lines and the ruins of the most advanced American aircraft ever built adorning the walls of a Serbian museum, and this was 15 years ago. How far do you think radars have come since?

IS STEALTH DEAD

Whatever we may feel about stealth technology and its shortcomings, the technology still has its place in military operations and if it wasn't effective, the world's mightiest armed forces will not be vying for it.

It's entirely possible that we're correct in thinking what we're thinking, but we're only civilians and the true potential of any technology in any military application is something that the military themselves are aware of. There might be some self-delusion involved, ideologies being bandied about, but we just have to assume, or rather, hope that they know best.

The F-35 fiasco suggests otherwise of course, but you never know. Within 10 years, the F-35 might be the most fearsome fighter there is. Then again, it might be the biggest failure of all time.